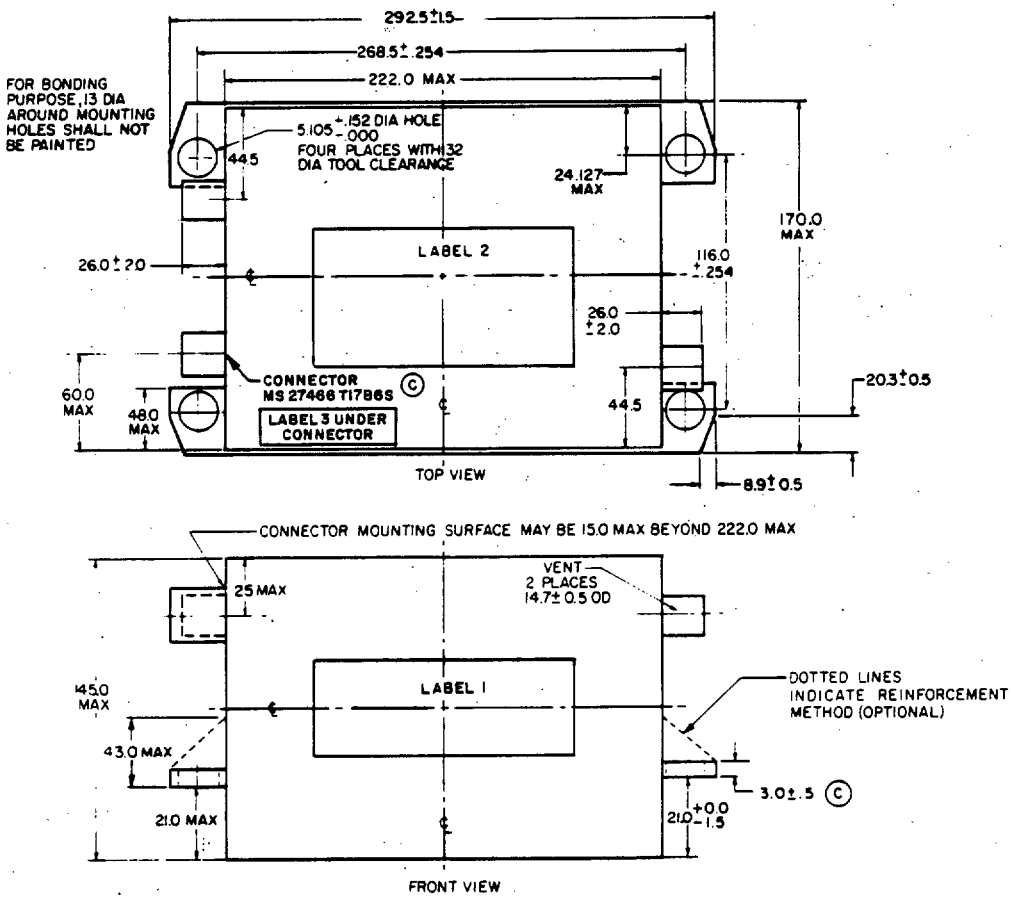
SPECIFICATION SHEET

Ⓒ BATTERY, STORAGE, AIRCRAFT, MEDIUM-RATE, TYPE 1, MAINTENANCE-FREE, 24-VOLT, 7.5 AMPERE-HOUR

This specification sheet is approved for use by the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the storage battery described herein shall consist of this document and the latest issue of Specification MIL-B-8565.



NOTES: 1. Tolerances not shown shall be 3.0.
2. All dimensions are in millimeters.

FIGURE 1. Dimensions P/N D8565/4-1.

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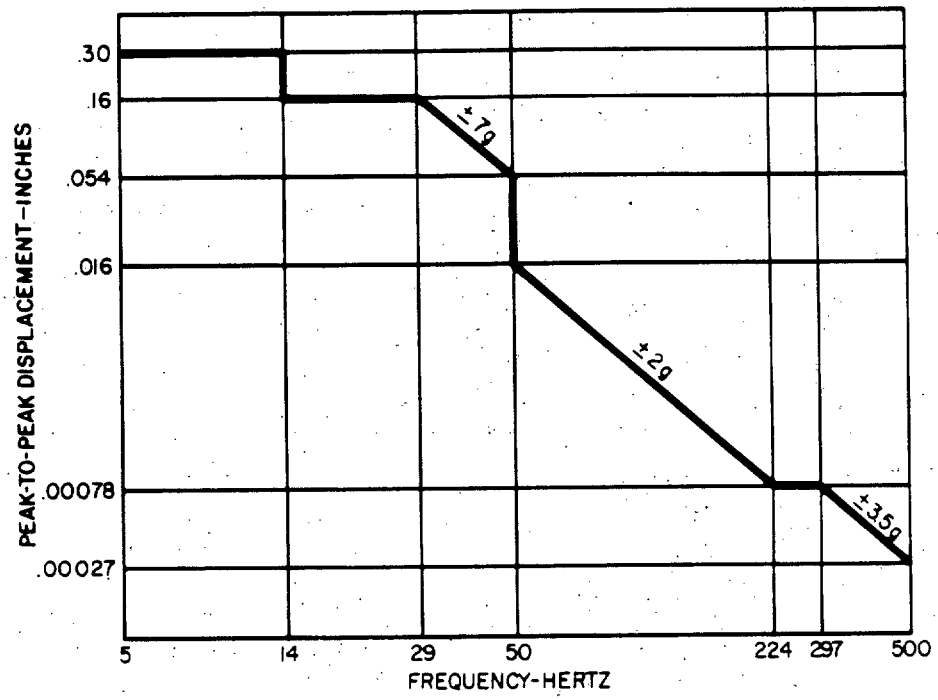


FIGURE 2. Sine wave vibration test levels (4.6.23).

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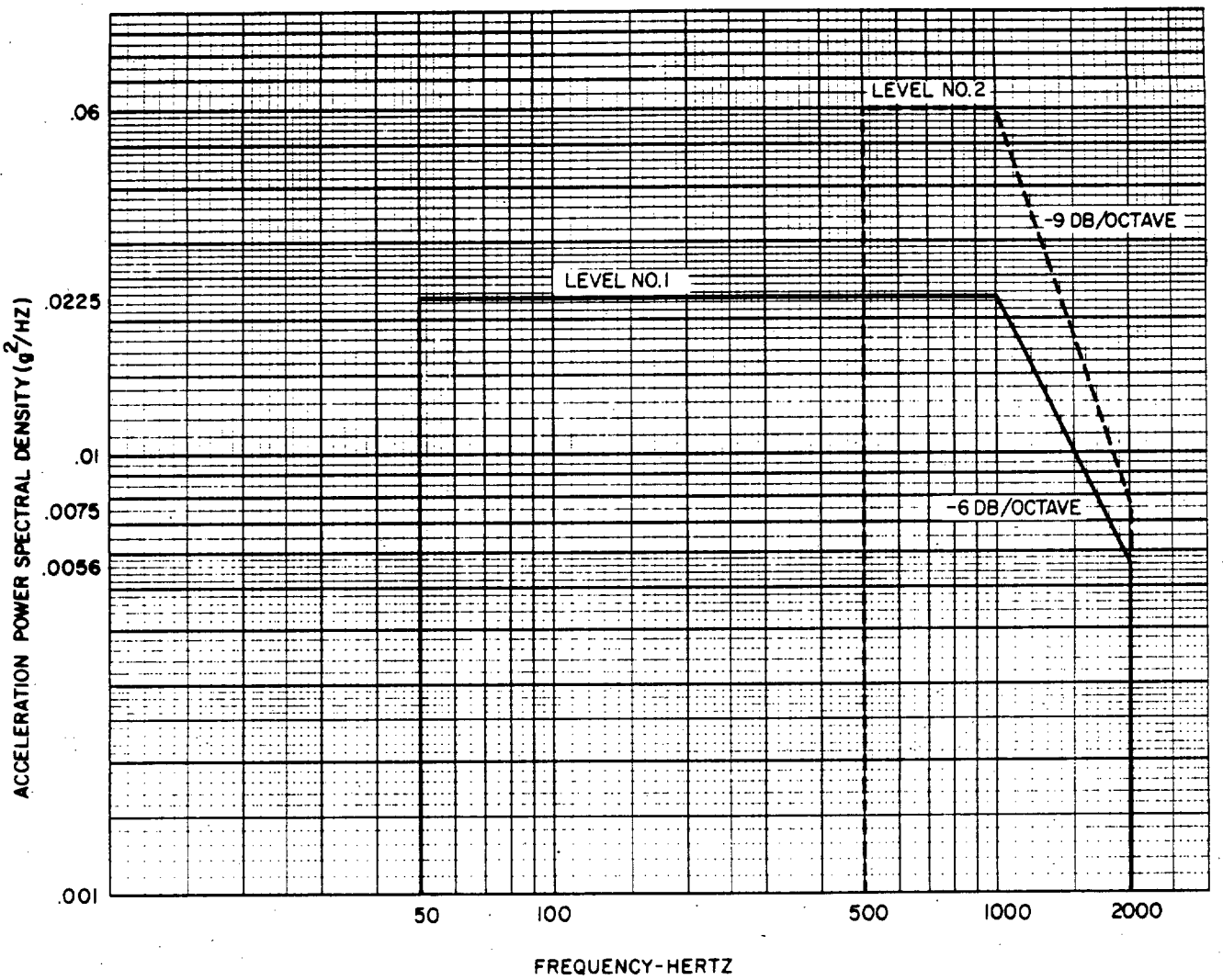


FIGURE 3. Random vibration test levels (4.6.23).

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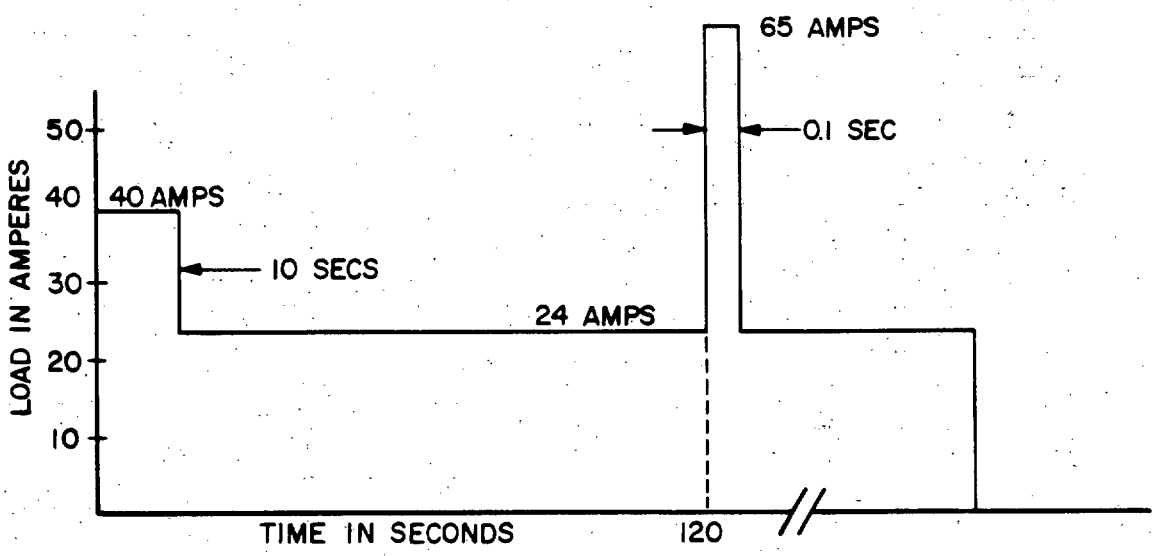


FIGURE 4. Emergency loads (3.6.23.1 and 4.6.20.1.1).

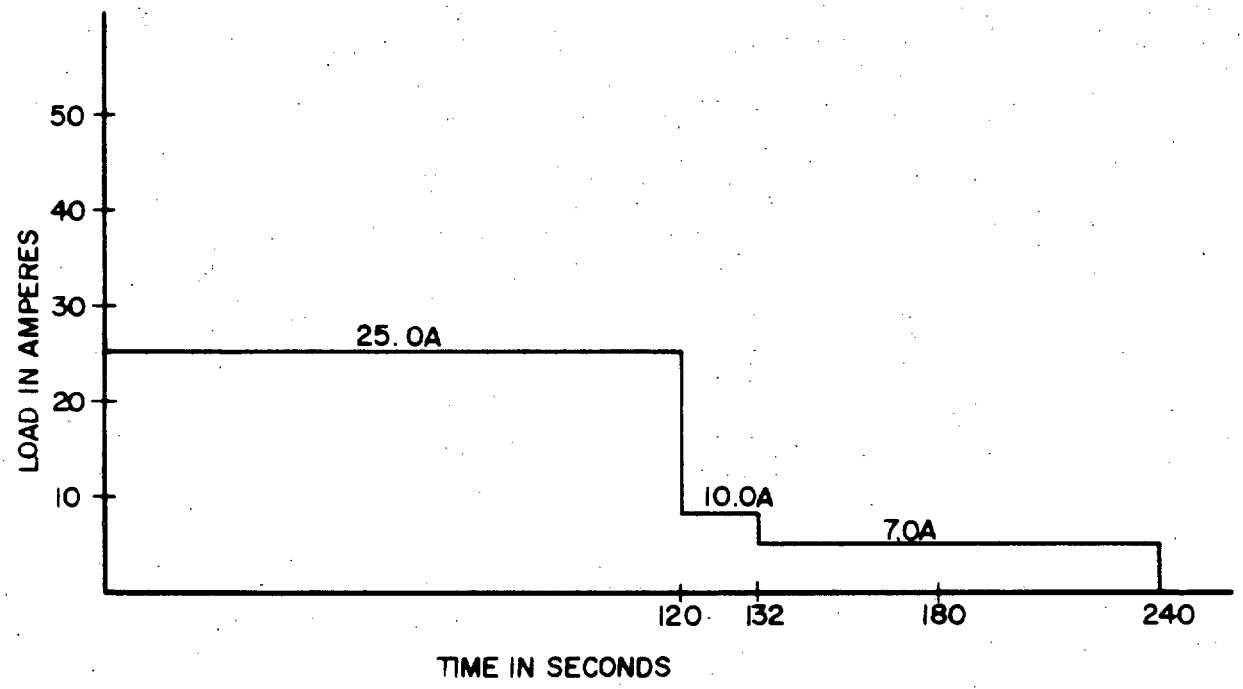


FIGURE 5. Start-up loads (3.6.23.2 and 4.6.20.1.2).

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REQUIREMENTS:

1. The dimensions of the battery shall be as shown in figure 1.

2. The military part number of the battery shall be D8565/4-1.

3. The rated capacity of the battery shall be 7.5 ampere-hours (7.5 Ah/1 HR/21°C/18.0V at end of life). 1C = 7.5 amperes.

4. The weight of the battery shall not exceed 11.8 kg.

(C) 5. Aircraft installation. For installation in the aircraft, the battery shall be supported as shown in figure 1 of MIL-B-8565. Vibration isolation shall be provided to insure that the vibration level does not exceed the requirements of the 2-year accelerated life tests of 4.6.13. If vibration isolators are used, sufficient sway clearance between the battery and the mounting structure shall be provided.

(C) 6. The battery shall conform to military specification MIL-B-8565 for Type 1 batteries with the following additions and deletions.

3.5.5 Receptacles. Add: "The receptacle shall be MS27466T17B6S.

a. Pin A shall be high side AC (input heater power).

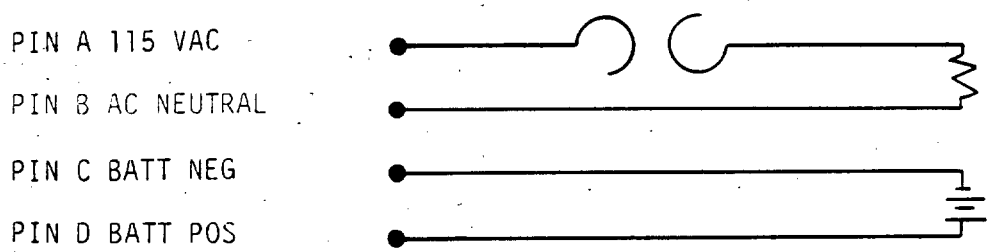
b. Pin B shall be AC ground (heater return).

c. Pin C shall be battery negative.

d. Pin D shall be battery positive."

(C) 3.6.3.2.1 Polarity marking. Delete and add the following: "On the battery surface that contains the connector, the following shall be marked with lettering height of .15 inch minimum:"

CONNECTOR PIN IDENTIFICATION



If heaters are not incorporated in the battery, the words "NO HEATER USED" shall be substituted in place of the heater circuit."

(C) 3.6.8 Capacity and electrical performance. Delete "Requirement Number (4)."

(C) 3.6.9 Strength of receptacle. Delete.

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Add the following paragraphs:

3.6.23 Battery output performance.

(C) 3.6.23.1 Emergency loads. Batteries, when discharged in accordance with the load profile sequence shown below shall maintain a terminal potential of 18 volts or greater.

(Ref. Figure 4)

Step	Disch. Temp. (°C)	Load Current (A)	Discharge Time (Seconds)
1	25	40 24 65 24	10 110 0.1 480
2	-29	40 24 65 24	10 110 0.1 50
3	-18	40 24 65 24	10 110 0.1 180
4	0	40 24 65 24	10 110 0.1 300
5	55	40 24 65 24	10 110 0.1 480

3.6.23.2 Start-up load. The battery, when discharged in accordance with figure 5 at temperatures in the range of -29.0°C to 55°C, shall maintain a terminal potential of 18 volts or greater.

3.6.24 Cold and hot temperature charging.

(C) 3.6.24.1 Half-hour charging. The battery having a 3.5 Ah capacity at temperatures in the range of -18°C to 55°C shall be capable of delivering 7.0 Ah capacity after a 28.25 ± 0.5V charging source has been applied to the battery for 30 minutes.

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(C) 3.6.24.2 One-hour charging. The battery, when discharged at temperatures in the range of -40°C to -17°C at 7.5A to 18.0V, shall be capable of delivering 7.0 Ah capacity after a 28.25 ± 0.5V charging source has been applied to the battery for 1.0 hour. The battery shall not be damaged when a 28.25 ± 0.5V charging source is applied continuously to the battery at -40°C.

3.6.24.3 Heaters. Heaters may be incorporated into the battery to meet the cold temperature charging requirements. The heaters shall utilize 115 VAC power in accordance with MIL-STD-704, accessible at connector pin A. Heater current shall not exceed 3A at 115V. Heater power shall not be provided during any discharge performance in accordance with this specification except during vibration. Heater power shall be provided during all constant-potential charges.

3.6.24.3.1 The heater element shall be fabricated and installed in the battery assembly using insulated heaters and wire assemblies so as to minimize the possibility of sparking.

3.6.24.3.2 The battery shall contain temperature limiting devices as needed to prevent overheating when the heater circuit is energized. Under any condition of operation specified herein, the battery case temperature shall not exceed 75°C due to heater and battery operation. If the battery case temperature exceeds 49°C, caution labels stating: "CAUTION HOT SURFACE" shall be attached to the battery cover and to the surface bearing the battery identification. Both temperature conditions above apply at an ambient temperature of 25° ± 5°C. The heater control devices shall be compatible with the battery heater current requirements. If thermal switches are used, they shall be a snap-action type, in accordance with MIL-S-24236. The switches shall be designed and installed to prevent ignition of an explosive mixture of hydrogen and oxygen. All components of the heater circuit shall be located inside the battery case. In the event of primary heater control failure, the battery shall not reach a damaging temperature.

(C) 4.6.11 Constant voltage discharge (14.0 volts). Delete.

(C) 4.6.12 Strength of receptacle. Delete.

4.6.13 Life. Change the number of cycles for the medium-rate, type I battery from 85 to 40.

(C) 4.6.15 Charge and discharge test at low temperature. Delete.

4.6.20 Special tests. Add:

4.6.20.1 Battery output performance test.

(C) 4.6.20.1.1 Emergency loads. Battery samples 1, 2 and 3 shall be used to demonstrate the performance requirements of 3.6.23.1. The tests shall be conducted as follows:

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a. Discharge samples 1, 2 and 3 in accordance with step 1 of 3.6.23.1.

b. Charge each battery at 28.25 ± 0.5V for 2 hours at 25°C.

c. Discharge samples as follows:

Sample 1 in accordance with step 2 of 3.6.23.1.

Sample 2 in accordance with step 3 of 3.6.23.1.

Sample 3 in accordance with step 4 of 3.6.23.1.

d. Charge each battery at 28.25 ± 0.5V for 2 hours at 25°C.

e. Discharge sample 1 in accordance with step 5 of 3.6.23.1.

f. Charge sample 1 at 28.25 ± 0.5V for 2 hours at 25°C.

4.6.20.1.2 Start-up loads. Battery samples 1, 2 and 3 shall be used to demonstrate the discharge performance requirements of 3.6.23.2. The test shall be conducted as follows:

a. Soak battery sample 1 at 55°C, sample 2 at -29°C and sample 3 at 25°C for 12 hours.

b. While each battery remains at these temperatures, discharge the batteries in accordance with figure 5.

4.6.20.2 Cold and hot temperature charge test.

(C) 4.6.20.2.1 Half-hour charging. Battery samples 1, 2 and 3 shall be used to demonstrate the cold and hot temperature charging requirements of 3.6.24. The test shall be conducted as follows.

a. Charge each battery at 28.25 ± 0.5V for 5 hours at 25°C.

b. Stabilize sample 1 at -18°C, sample 2 at 15°C and sample 3 at 55°C, then discharge at 7.5A to 18.0V. Each battery shall provide at least 3.5 ampere-hours discharge.

c. Stabilize each battery at 25°C, then charge at 28.25 ± 0.5V for 2 hours.

d. Stabilize each battery at -18°C, 15°C and 55°C, respectively, then discharge at 7.5A to remove all but 3.5 Ah from batteries, as determined by 4.6.20.2.1b.

e. Rest battery to stabilize at test temperatures, then apply 28.25 ± 0.5V for 30 minutes.

f. Immediately following charge, remove batteries from test chambers and discharge at 7.5A to 18.0V. Each battery must deliver 7.0 Ah minimum.

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Ⓒ 4.6.20.2.2 One-hour charging. Battery sample 3 shall be subjected to a -40°C charge demonstration test and shall be conducted as follows:

a. Charge the battery at 28.25 ± 0.5V for 2 hours at 25°C.

b. Stabilize the battery at -40°C, and then discharge at 7.5A to 18.0V.

c. Rest battery to stabilize at -40°, then apply 28.25 ± 0.5V for 1.0 hour.

d. Immediately following charge, remove battery from test chamber and discharge at 7.5A to 18.0V. Battery shall deliver 7.0 Ah minimum.

Ⓒ 4.6.23 Vibration. Delete and add the following: "The battery assembly, including connector and cable, shall be mounted to the test apparatus, per 4.5.5, and vibrated along each orthogonal axis, per figure 1, in accordance with the test levels of figure 2 of the specification sheet. Two 15-minute logarithmic sweeps shall be conducted from 5-500-5 Hertz and one 5-minute logarithmic sweep will be conducted from 20-30-20 Hertz. A resonance search shall be conducted from 5-500 Hertz and resonance dwell testing performed for 1.5 minutes at each resonance, to a maximum of three most severe exhibiting a transmissibility of two or greater. The battery shall also be subjected to two separate random vibration tests along each orthogonal axis; one for a period of 20 minutes per axis to Level No. 1 of figure 3 of the specification sheet, and the second for a period of 5 minutes per axis to Level No. 2 of figure 3 of the specification sheet. The battery heaters shall be energized, and the battery charged at 28.25 ± .5 volts during vibration, and shall be discharged at 25.0A for 2 minutes prior to beginning each axis of vibration."

Figure 2 of MIL-B-8565. Delete.

Preparing Activity:
Navy - AS
Project No. 6140-N649

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TABLE 1. Inspection of batteries and order of test.

Inspection Number	Inspections	Qualification Inspection			Requirements Paragraph	Method of Inspection Paragraph	Quality Conformance Inspection		
		Qualification Sample Number					Group A Tests	Group B Tests	
		1	2	3				1	2
1	Visual & Mechanical Examination	X	X	X	3.5, 3.6.2.2 & 3.6.3	4.6.3	X	X	X
2	Dimensions & Weight	X	X	X	3.6.5	4.6.4	X	X	
3	Dielectric Strength	X	X	X	3.6.4	4.6.5	X	X	X
4	Strength of Vent Tubes	X	X	X	3.6.7	4.6.7	X	X	X
5	Color and Marking	X	X	X	3.6.3	4.6.8	X	X	X
6	Conditioning Charge	X	X	X	4.5.1.4	4.6.9		X	X
7	Capacity Discharge	X	X	X	3.6.8	4.6.10	X	X	X
8	Battery Output Performance	X	X	X	3.6.23	4.6.20.1			
9	Cold Temperature Charge	X	X	X	3.6.24	4.6.20.2			
10	Life Cycling	X			3.6.10	4.6.13		X	
11	Capacity Discharge at 49°C		X		3.6.8	4.6.14			
12	Discharge While Inverted		X		3.5.8 & 3.6.8	4.6.16			X
13	Altitude		X		3.6.11 & 3.6.12	4.6.17			X
14	Mechanical Shock	X			3.6.11 & 3.6.12	4.6.18			
15	Temperature Shock	X			3.6.11 & 3.6.12	4.6.19		X	
16	Temperature Rise & Float		X		3.6.13	4.6.21			X
17	Battery Gas Emission Test		X		3.6.14	4.6.22			
18	Vibration		X		3.6.16	4.6.23			X
19	Humidity		X		3.6.17	4.6.24			X
20	Salt Fog	X			3.6.18	4.6.25		X	
21	Ground Storage			X	3.6.15	4.6.26			
22	Physical Integrity at High Temperature	X			3.6.12	4.6.27			
23	Final Examination	X	X	X	3.6.4	4.6.30		X	X

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